you would expect for this counter.

- Select an optimal operating voltage and LLD based on previous measurements so that all source counts can be recorded
- Take a 1 minute measurement and use the count rate to determine the current activity as well as the original source activity.
  - Use the date marked on the source to calculate the original activity
  - Compare your calculated rate with the original source activity.
  - How should the rate you measure be related to the marked activity?

4 Gamma Response

Observe the counter response to gamma rays only by using the ^{60}Co or ^{137}Cs source turned upside down. In this manner, the betas are blocked, but the higher energy gamma rays will penetrate into the counter gas.

- Repeat the measurements described in Sections 2.2.3 and 2.3.2
  - Again, for this section do not record the centroid channel (no peak will be visible).

5 Lab Report Requirement

Section 2.2.2

- Plot the peak centroid vs. the shaping time. Calculate the ballistic deficit at 0.5 μs relative to that at 10 μs in terms of percent. Discuss the evolution of the rise time with respect to the shaping time of the amplifier output pulse.

Section 2.3.1

- Plot the counting curve, and discuss whether the shape of the counting curves meet expectations.
- Plot the amplitude vs. voltage on a log-linear scale. Compare with the variation in gas multiplication shown in Fig. 6.11 of Knoll.
- Identify all prominent features in the differential pulse height spectrum.
- Perform channel-energy calibration for the spectrum assuming that the relationship is linear. Make sure you include all the experimental settings in your plot legend (amplifier gain, bias on the detector, etc.). Show the calibrated energy spectrum of the alpha source.

Section 2.3.2

- Plot the counting curve, and discuss whether the shape of the counting curves meet expectations.
- Discuss the relative amplitudes of beta and alpha pulse heights.

Section 3

- Calculate the absolute efficiency and intrinsic efficiency of the detector. Discuss whether the intrinsic efficiency is as expected, and if not discuss sources of error.

Section 4

- Plot the counting curve, and discuss whether the shape of the counting curves meet expectations.